Skills Summary

Adding, Subtracting, and Multiplying Complex Numbers

Use this reference while completing your worksheet or when studying.

Treat i as a variable while you distribute and collect, then — and only then — use the one fact that makes complex numbers different: $i^2 = -1$. Every answer belongs in standard form $a + bi$.

1. Adding and subtracting

Rule: combine like parts.

$$(a + bi) \pm (c + di) = (a \pm c) + (b \pm d)i$$

Subtraction distributes the minus sign over *both* parts of the second number. Nothing here needs i^2; addition and subtraction never create one.

Example: $(6 + 2i) + (1 - 9i)$
Step 1. No products appear, so this is pure collecting — real with real, imaginary with imaginary.
Step 2. Real: $6 + 1 = 7$; imaginary: $2i - 9i = -7i$.
$\boldsymbol{7 - 7i}$

Try it: $(2 - 5i) - (7 + 3i)$

check: $\boldsymbol{-5 - 8i}$

2. Multiplying complex numbers

Rule: distribute all four products, then replace i^2 with -1 and re-collect.

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i$$

Conjugates: $(a + bi)(a - bi) = a^2 + b^2$, always real — the outer and inner products cancel and the $-b^2i^2$ becomes $+b^2$.

Example: $(3 + 4i)(2 - i)$
Step 1. Two binomials, so FOIL first and postpone every simplification until all four products are on the page.
Step 2. $6 - 3i + 8i - 4i^2 = 6 + 5i + 4$, using $-4i^2 = +4$.
$\boldsymbol{10 + 5i}$

Try it: $(1 + 6i)(2 + 3i)$

check: $\boldsymbol{-16 + 15i}$

3. Powers of i

Rule: the powers cycle with period 4.

$$i^1 = i, \quad i^2 = -1, \quad i^3 = -i, \quad i^4 = 1$$

For i^n, divide n by 4 and keep only the remainder r; then $i^n = i^r$, with $i^0 = 1$.

Example: Simplify i^{27}.
Step 1. The exponent is large but the powers repeat every four, so only the remainder on division by 4 matters.
Step 2. $27 = 4 \cdot 6 + 3$, remainder 3, so $i^{27} = i^3$.
$\boldsymbol{-i}$

Try it: Simplify i^{30}.

check: $\mathbf{-1}$

4. Complex solutions of a quadratic

Rule: when $\Delta = b^2 - 4ac < 0$, write $\sqrt{\Delta} = i\sqrt{|\Delta|}$ and finish the quadratic formula as usual.

$$x = \frac{-b \pm i\sqrt{|\Delta|}}{2a}$$

The two solutions are always conjugates, so checking one checks both.

Example: Solve $x^2 - 4x + 5 = 0$.
Step 1. $\Delta = 16 - 20 = -4$ is negative, so there are no real solutions and the square root must be written with i.
Step 2. $x = \dfrac{4 \pm i\sqrt{4}}{2} = \dfrac{4 \pm 2i}{2}$, then divide both parts by 2.
$\boldsymbol{x = 2 + i}$ or $\boldsymbol{x = 2 - i}$

Try it: Solve $x^2 + 2x + 10 = 0$.

check: $\boldsymbol{x = -1 + 3i}$ or $\boldsymbol{x = -1 - 3i}$